Annex-B

FYP SRS Document

Final Year Project

Software Requirement Specification

For

Detection of Invasive Ductal Carcinoma (IDC)

Bachelors of Computer Sciences (BSCS)

By

S#	Name	Registration #/Roll #/Section	Mobile #	E-Mail
1.	Muhammad Uzair Ishfaq	SP-2020/BSCS/030/A	+92 3154784468	Sp20-bscs-030-@lgu.edu.pk
2.	Khadija Ibrahim	SP-2020/BSCS/011/A	+92 312 4301367	Sp20-bscs-011-@lgu.edu.pk

Supervised by:

Teacher Name ___________________ (Signature)

Department of Computer Science

Lahore Garrison University

Lahore

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1. Introduction

This document is Software Requirements and Specifications (SRS) for **Detection of Invasive Ductal Carcinoma (IDC)**. This is the Final Year Project of above-mentioned student for session SP-2020-BSCS-A. It is prepared by keeping the pattern provided by university for SRS.
Our project is aiming for better utilization of Artificial intelligence for developing DL based automated system that can **detect invasive ductal carcinoma in breast histopathology images**.

1.1 Purpose

The software requirements specified in this document serve the purpose of outlining the development and implementation of a DL-based system for the accurate and efficient **detection of Invasive Ductal Carcinoma (IDC) in breast histopathology images**. This Software Requirements Specification (SRS) is intended to guide the development team in creating a robust and user-friendly web application that aids medical professionals in diagnosing and treating breast cancer at an early stage.

The scope of the product covered by this SRS includes the entire development process, from data collection and preprocessing to algorithm development, model training, and the final deployment of the web application. It also encompasses the integration of cutting-edge deep learning techniques and the creation of a comprehensive documentation package for the system.

The primary purpose of this project is to address the limitations and challenges associated with manual interpretation of histopathology images, such as inter-observer variability and potential diagnostic errors. By leveraging advanced deep learning methodologies, the proposed system aims to achieve a high level of accuracy in the detection of **IDC**, thereby facilitating timely and accurate medical interventions for patients. Furthermore, the development of a user-friendly web application interface will ensure seamless accessibility and interpretation of results for healthcare professionals. Ultimately, this project aims to contribute to the advancement of medical technology and improve the standard of care for individuals diagnosed with breast carcinoma, thereby reducing the mortality rate associated with this prevalent form of cancer among women.

The document serves as a comprehensive reference for all stakeholders involved in the project, including developers, data scientists, medical professionals, and project managers. It acts as a blueprint for the system's design, development, and testing phases, ensuring a clear understanding of the project's objectives and constraints.

The document's specific purposes include:

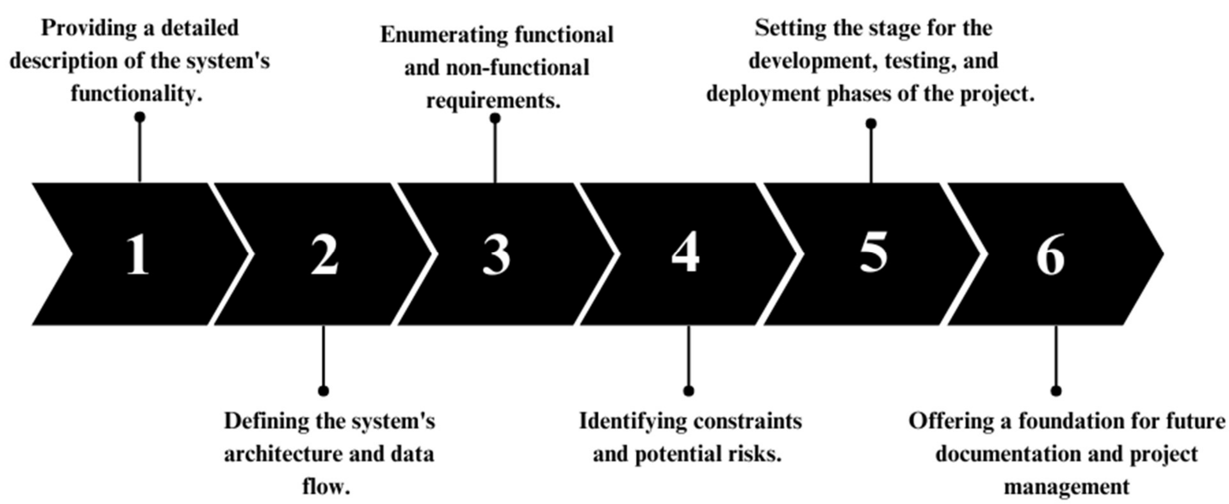

Figure 1 : Document Specific Purposes

1.2 Document Conventions

This section outlines the conventions and notations used throughout this Software Requirements Specification (SRS) document to facilitate understanding and clarity.

The following conventions are employed:

1. Terminology:

- ✓ **IDC:** Invasive Ductal Carcinoma, a common subtype of breast cancer.
- ✓ **DL:** Deep Learning, a subfield of artificial intelligence.
- ✓ **GPU:** Graphics Processing Unit, a hardware component often used for accelerating deep learning tasks.
- ✓ **CNN:** Convolutional Neural Network, a type of deep learning model commonly used for image analysis.
- ✓ **WebApp:** Web Application, a user interface accessible through web browsers.
- ✓ **SRS:** Software Requirements Specification, the document being referred to.

2. Formatting:

- ✓ **Bold Text:** Key headings and subheadings are in bold for easy identification.
- ✓ **Monospaced Text:** Used for code snippets, file names, and technical terms.
- ✓ **Hyperlinks:** Links to external resources or references are enclosed in square brackets.

3. Lists:

- ✓ **Numbered Lists:** Used for sequential steps or items that should be performed in a specific order.
- ✓ **Bullet Points:** Used for items that do not require a specific sequence.

4. Figures and Tables:

- ✓ **Figures:** (e.g., Figure 1, Figure 2) are used to visually represent diagrams, charts, and images.
- ✓ **Tables:** (e.g., Table 1, Table 2) are used for presenting structured data.

5. References:

- ✓ Citing external sources and references follows the format [Author, Year etc.], **e.g.**, [1] for the first reference, [2] for the second, and so on. A complete list of references is provided at the end of this document.

6. Acronyms and Abbreviations:

- ✓ Acronyms and abbreviations are spelled out upon their first usage with the acronym or abbreviation provided in parentheses. **For example**, "Deep Learning (DL)".

These document conventions aim to enhance the readability and comprehension of the **SRS**, ensuring that all stakeholders can understand and interpret the content effectively.

1.3 Intended Audience and Reading Suggestions

This SRS document is intended to be read and referenced by a diverse group of stakeholders, including but not limited to:

- ✓ **FYP Supervisor**
- ✓ **FYP Team**
- ✓ **Developers:** Those responsible for designing, developing, and implementing the IDC breast cancer detection system.

- ✓ **Data Scientists:** Professionals involved in the development of deep learning algorithms, model training, and optimization.

- ✓ **Medical Professionals:** Healthcare providers and experts in breast cancer pathology who will use the system for diagnostic purposes.

- ✓ **Project Managers**: Individuals responsible for project planning, scheduling, and coordination.

- ✓ **Quality Assurance/Testers**: Teams responsible for testing and validating the system's functionality and performance.

- ✓ **Regulatory Authorities:** Entities responsible for ensuring that the system complies with medical and data privacy regulations.

- ✓ **Investors and Stakeholders:** Individuals or organizations with a financial or strategic interest in the project's success.

By addressing the needs and expectations of this diverse audience, this document aims to provide a comprehensive and clear understanding of the **IDC** breast cancer detection system's requirements and specifications.

1.4 Product Scope

The software specified in this document is a Deep Learning (DL)-based system designed for the accurate and efficient **detection of Invasive Ductal Carcinoma (IDC) in breast histopathology images**. This system aims to revolutionize the process of breast cancer diagnosis, specifically addressing the challenges associated with manual interpretation of histopathology images, which are prone to errors and subject to inter-observer variability.

The purpose of this software is to provide medical professionals with a reliable and automated tool that can swiftly and accurately detect IDC, enabling timely and effective intervention for patients diagnosed with breast carcinoma. By leveraging advanced DL techniques, the software seeks to enhance the accuracy of diagnosis and improve patient outcomes by enabling early detection and timely treatment.

Relevant benefits of this software include:

- ✓ **Enhanced Accuracy:** The system is designed to surpass existing standards in IDC detection accuracy, minimizing the risk of misdiagnosis and ensuring precise and reliable results.

- ✓ **Efficiency:** By automating the detection process, the software enables medical professionals to make prompt decisions, leading to early intervention and improved patient prognosis.

- ✓ **Improved Patient Outcomes:** Early detection of IDC can significantly improve treatment success rates and reduce mortality associated with late-stage breast cancer, ultimately improving patient outcomes and quality of life.

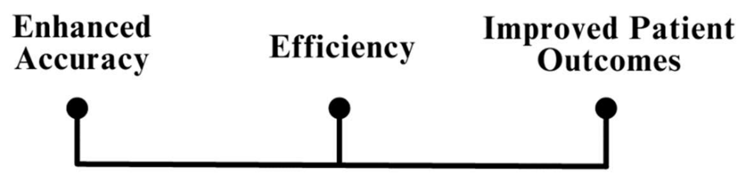

Figure 2: Benefits of this software

2. Overall Description

The IDC breast cancer detection system is a software application designed to aid medical professionals in the early and accurate detection of **Invasive Ductal Carcinoma (IDC) in breast cancer pathology images**. This system leverages deep learning algorithms to analyze medical images, assess the presence of IDC, and provide diagnostic results.

Key Components of the System:

- ✓ **Image Data Input:** This component allows users, primarily medical professionals, to upload breast cancer pathology images for analysis. These images can be in various formats, and the system ensures data integrity and security during the upload process.

- ✓ **Preprocessing:** Once the images are uploaded, the system applies a series of preprocessing techniques to enhance their quality. These techniques include normalization, resizing, and noise reduction. High-quality input data is critical for accurate model predictions.

- ✓ **Deep Learning Model Training:** The system incorporates state-of-the-art deep learning algorithms to train a model for IDC detection. Training is performed using labeled datasets that contain examples of both IDC-positive and IDC-negative cases. The goal is to teach the model to accurately classify images as containing IDC or not.

- ✓ **Inference and Diagnosis:** After model training, the system is capable of performing inference on new, unseen images. It utilizes the trained model to predict whether IDC is present in the images. This component provides diagnostic results, aiding medical professionals in making informed decisions about patient care.

- ✓ **Reporting:** The system generates diagnostic reports that include the results of the IDC detection process. These reports are designed for medical professionals and provide essential information about the presence or absence of IDC in the analyzed images. The IDC breast cancer detection system aims to improve the accuracy and efficiency of breast cancer diagnosis, potentially leading to earlier detection.

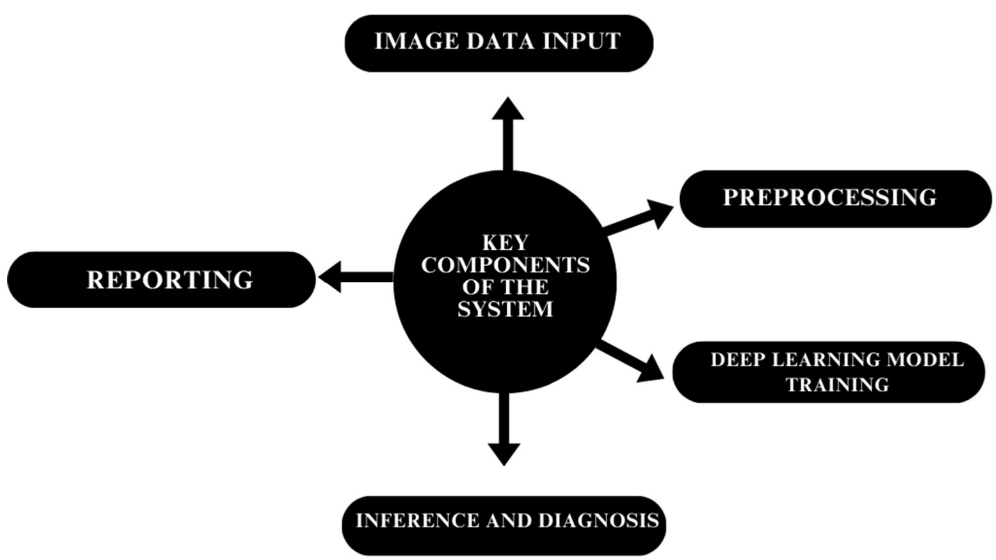

Figure 3: Key Components of the System

2.1 Product Perspective

The software system specified in this Software Requirements Specification (SRS) is a new, self-contained product developed as a standalone solution for the automated detection of Invasive Ductal Carcinoma (IDC) in breast histopathology images. It is not directly tied to any existing product line or part of a larger system but serves as an innovative addition to the healthcare technology landscape, specifically focused on addressing the challenges associated with manual interpretation of breast cancer pathology images.

Contextually, the software system operates within the domain of healthcare, targeting the critical need for accurate and timely diagnosis of breast cancer. As a standalone product, it integrates seamlessly with existing healthcare infrastructure, enabling medical professionals to leverage its capabilities for improved decision-making and patient care. While it does not directly replace any existing systems, it complements the current diagnostic procedures by providing an automated and reliable method for IDC detection, thereby enhancing the efficiency and accuracy of breast cancer diagnosis.

The major components of the overall system include the following:

- ✓ **DL-based Algorithm Module:** This module forms the core of the system, incorporating sophisticated deep learning techniques for the automated analysis of histopathology images and accurate detection of IDC.
- ✓ **Web Application Interface:** Serving as the primary interaction point for users, this interface allows medical professionals to upload histopathology images, receive IDC detection results, and interpret the findings for patient diagnosis and treatment. The diagram below illustrates the major components of the system:

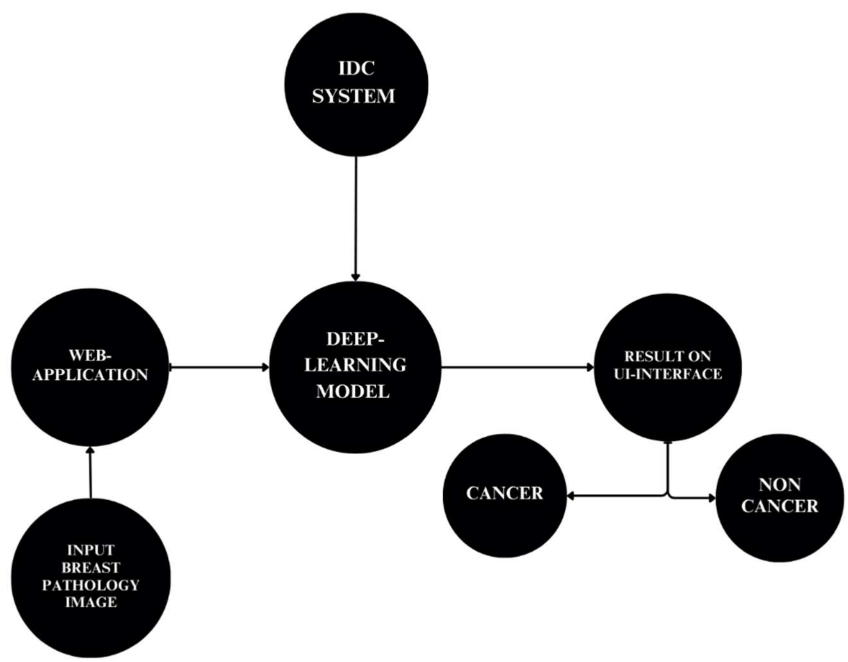

Figure 4: Components of the overall system

2.2 Product Functions

Functional Requirements define the specific features and capabilities of the system:

1. **Image Data Input:**
   - ✓ The system shall allow authenticated users to upload breast cancer pathology images.
   - ✓ The system shall support various image formats, including but not limited to JPEG, PNG, and DICOM.

- ✓ The system shall verify the integrity and authenticity of uploaded images.

2. **Preprocessing:**
   - ✓ The system shall apply image preprocessing techniques, including normalization, resizing, and noise reduction, to enhance data quality.
   - ✓ Preprocessed images shall be stored for model training and inference.

3. **Deep Learning Model Training:**
   - ✓ The system shall utilize deep learning algorithms to train a model for IDC detection.
   - ✓ The training process shall use labeled datasets that contain examples of **IDC-positive and IDC-negative cases.**
   - ✓ The trained model shall be saved for future inference.

- **Inference and Diagnosis:**
  - ✓ The system shall perform inference on new, unseen images using the trained model.
  - ✓ Inference shall result in a binary classification **(IDC-positive or IDC negative)** for each image.
  - ✓ Inference results shall be stored for reporting.

- **Reporting:**
  - ✓ The system shall generate diagnostic reports for medical professionals.
  - ✓ Diagnostic reports shall include the results of IDC detection, including the confidence level of the predictions.
  - ✓ Reports shall be accessible to authorized users.

2.3 User Classes and Characteristics

The software system is intended for use by various user classes within the healthcare domain, particularly those involved in the diagnosis and treatment of breast cancer.

The following user classes are anticipated to interact with the product:

- ✓ **Oncologists:**
  - ✓ **Characteristics:** Highly experienced medical professionals specializing in the diagnosis and treatment of cancer. They possess in-depth knowledge of cancer pathology and are responsible for making critical decisions regarding patient treatment plans.

  - ✓ **Usage:** Oncologists are primary users of the system, relying on its accurate IDC detection capabilities to make informed decisions about patient care and treatment strategies.

- ✓ **Pathologists:**
  - ✓ **Characteristics:** Medical professionals with expertise in analyzing and interpreting histopathology images. They play a crucial role in identifying cancerous cells and providing accurate diagnoses.
  - ✓ **Usage:** Pathologists utilize the software to enhance the efficiency and accuracy of their diagnostic processes, enabling them to identify and analyze potential IDC cases more effectively.

- ✓ **Radiologists:**
  - ✓ **Characteristics:** Experts in medical imaging and diagnostic techniques, particularly in the interpretation of medical images such as X-rays, CT scans, and MRI scans. They possess

the technical knowledge necessary for understanding and interpreting complex imaging data.

- ✓ **Usage:** Radiologists may utilize the software to complement their existing diagnostic procedures, particularly in cases where imaging data may indicate the presence of breast carcinoma.

- ✓ **Medical Researchers:**
  - ✓ **Characteristics:** Professionals engaged in research activities focused on the advancement of medical technology and treatment methods. They often require access to accurate diagnostic tools for conducting research studies and clinical trials.
  - ✓ **Usage:** Medical researchers may use the software system to analyze and evaluate its performance, contributing to the ongoing refinement and improvement of the IDC detection algorithms and methodologies.

- ✓ **Medical Students and Residents:**
  - ✓ **Characteristics:** Individuals undergoing medical education or residency programs, seeking to gain practical experience and knowledge in the field of oncology and pathology.
  - ✓ **Usage:** Medical students and residents may use the system for educational purposes, familiarizing themselves with the process of breast cancer diagnosis and the role of advanced technologies in improving patient care.

2.4 Operating Environment

The operating environment of this system is easy. To run higher management portal user need following Hardware:

- ✓ PC or Any Desktop or Any Laptop.
- ✓ Processor of Any company (Intel, AMD etc.).
- ✓ Display of Any screen.
- ✓ Network of Any good ISP provider.
- ✓ Operating System of Any OS (Windows, MacOS, Linux etc.).
- ✓ Browser of Any Updated Browser (Chrome, Safari, Opera, UC etc.).

2.5 Design and Implementation Constraints

The development and implementation of the software system for IDC detection in breast histopathology images are subject to several constraints that may influence the available options for the development team. These constraints include:

1. **Regulatory Compliance:**
   The software must adhere to all relevant healthcare regulatory policies and standards, ensuring patient data privacy and confidentiality in accordance with regional and international regulations, such as HIPAA (Health Insurance Portability and Accountability Act) and GDPR (General Data Protection Regulation).

2. **Hardware Limitations:**
   The system's performance may be influenced by the availability and capacity of hardware resources, particularly the need for high-performance computing, including Graphics Processing Units (GPUs) for efficient training and operation of the deep learning algorithms.

3. **Integration Requirements:**
   The software must be designed to seamlessly integrate with existing healthcare systems and infrastructures, ensuring compatibility with standard medical imaging technologies and data storage protocols.

4. **Data Quality and Standardization:**
   The effectiveness of the software system is highly dependent on the quality and standardization of the input histopathology image data. Therefore, ensuring the availability of a diverse and well-annotated dataset for training and validation purposes is critical.

5. **Security Considerations:**
   Data security and integrity are of utmost importance, necessitating the implementation of robust security protocols to protect sensitive patient information from unauthorized access or data breaches.

6. **Interoperability:**
   The system should be designed with interoperability in mind, allowing for seamless communication and data exchange between different healthcare applications and systems, facilitating the seamless flow of information across various medical platforms.
7. **Scalability:**
   The software should be designed to accommodate potential future expansions and enhancements, allowing for the integration of additional features and functionalities to meet the evolving needs of the healthcare industry.

8. **Performance Optimization:**
   The system should be optimized for high performance and efficiency, particularly in the processing and analysis of large-scale histopathology image datasets, ensuring timely and accurate results for healthcare professionals.

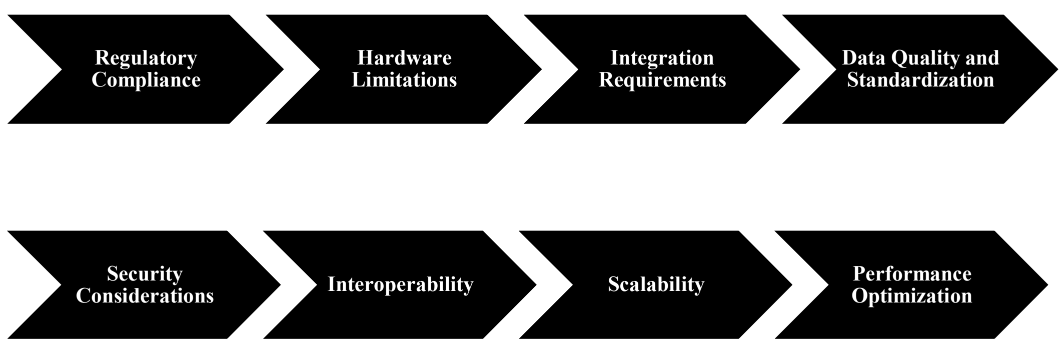

Figure 5: Design and Implementation Constraints

By recognizing and addressing these design and implementation constraints, the development team can ensure the successful deployment and operation of the software system, meeting the highest standards of quality, performance, and regulatory compliance within the healthcare industry.

2.6 User Documentation

The user documentation for the DL-based IDC detection system will be an essential part of our project to ensure that medical practitioners and users can effectively utilize the software.
The following components of user documentation will be delivered along with the software:

- ✓ **User Manual:** A comprehensive user manual will be provided, detailing how to use the web application for IDC detection. This manual will include step-by-step instructions, system requirements, and guidelines for uploading and interpreting histopathology images.

- ✓ **On-Screen Help:** The web application will include on-screen help features, including tooltips and context-sensitive help buttons, to assist users in real-time as they navigate the application.

- ✓ **Tutorials:** Interactive tutorials will be available within the web application to guide users through the process of uploading images, interpreting results, and understanding the IDC detection process.

- ✓ **FAQ Section:** A Frequently Asked Questions (FAQ) section will be included in the web application, addressing common queries and providing quick answers to user concerns.

- ✓ **Technical Specifications:** Users will have access to technical specifications, which will include information about system requirements, supported browsers, and other technical details related to the software.

- ✓ **Data Privacy and Security Guidelines:** Given the sensitive nature of medical data, the user documentation will outline the data privacy and security measures in place to protect patient information and ensure compliance with relevant regulations (e.g., HIPAA).

- ✓ **Troubleshooting Guide:** A troubleshooting guide will be included in the user documentation to help users address common issues they may encounter while using the software.

- ✓ **Contact Information:** Users will be provided with contact information for technical support and assistance, allowing them to reach out for help with any issues or questions.

2.7 Assumptions and Dependencies

Assumptions:

- ✓ **Data Availability:** We assume that a diverse and representative dataset of breast histopathology images will be available for training, validation, and testing. The accuracy of the deep learning model is highly dependent on the quality and diversity of the data.

- ✓ **Data Privacy and Compliance:** We assume that necessary consents and permissions for the use of patient data in the dataset have been obtained, and that the project complies with all relevant data privacy regulations and ethical standards.

- ✓ **High-Performance Computing Resources:** The project assumes access to high-performance computing resources, including GPUs, for efficient training of the deep learning model. The availability of these resources is essential to meet the project's efficiency objectives.

- ✓ **Web Development Tools:** The development of the web application relies on the availability and functionality of web development tools, frameworks, and libraries, as well as hosting infrastructure. We assume that these tools will be accessible and compatible with the project's requirements.

- ✓ **Stable Internet Connectivity**: Users will require a stable internet connection to use the web application. We assume that users will have access to reliable internet connectivity when using the system.

Dependencies:

- ✓ **Third-Party Libraries and Frameworks:** The project depends on the proper functioning of third-party libraries and frameworks for deep learning, image processing, and web development. Any changes or issues in these dependencies may impact the project.

- ✓ **Data Collection and Quality:** The availability and quality of the dataset used for training and validation are critical dependencies. If data collection is delayed or if the dataset is insufficient or of poor quality, it may affect the project's accuracy and efficiency goals.

- ✓ **Compliance and Ethics:** The project is dependent on the legal and ethical compliance of data collection and usage. Any changes in regulations or ethical considerations may require adjustments to the project's approach.

- ✓ **Hardware Resources:** Access to high-performance computing resources, such as GPUs, is essential for efficient model training. Any limitations or changes in the availability of these resources may impact the project's timeline and performance.

- ✓ **Web Hosting Services:** The web application's availability and performance depend on the web hosting services. Any issues related to hosting, including downtime or scalability, could affect user access and experience.

- ✓ **Internet Connectivity:** Users' ability to access the web application relies on their internet connectivity. If users face connectivity issues, it may impact their ability to use the system effectively.

- ✓ **Regulatory Approval:** If the project aims to be used in a healthcare setting, it may require regulatory approval. Any delays or challenges in obtaining such approval could impact the project's deployment.

- ✓ **Feedback from Medical Professionals:** The project's success is dependent on feedback and validation from medical professionals. Their timely input and feedback are crucial for improving the system and ensuring its accuracy and usability.

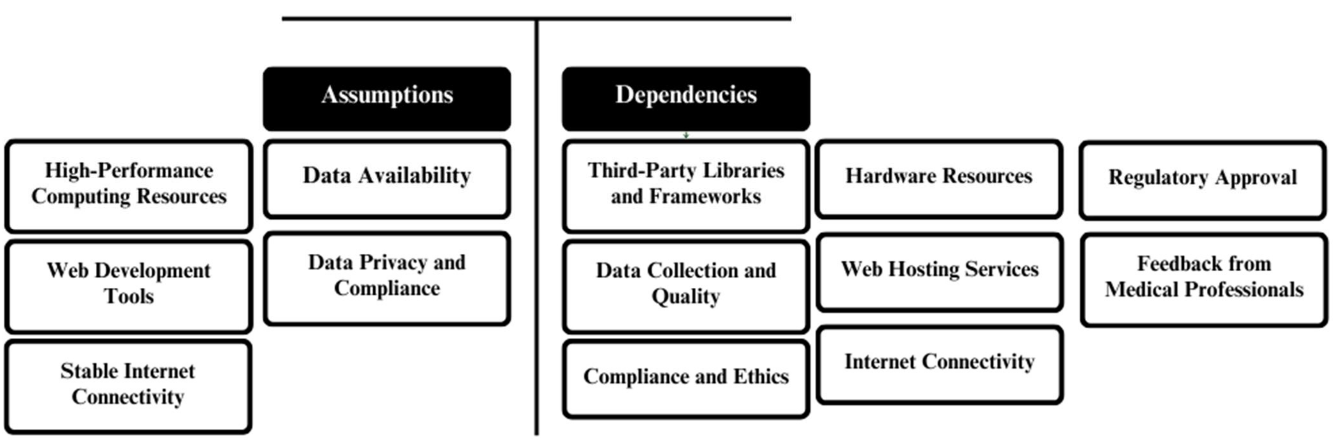

Figure 6: Assumptions and Dependencies

3. External Interface Requirements

3.1 User Interfaces

The software product will have a user-friendly web application interface designed to facilitate the efficient and accurate detection of Invasive Ductal Carcinoma (IDC) in breast histopathology images.

The logical characteristics of each interface between the software product and the users are as follows:

Software Components Requiring User Interface:

- ✓ **User Authentication and Access Control Component:** This interface will manage user authentication, ensuring secure access to the web application. It will include standard login and registration screens with appropriate input fields, error message display standards, and password recovery functionalities.

- ✓ **Image Upload Component:** This interface component will allow users to upload breast histopathology images for IDC detection. It will have a straightforward drag-and-drop or file selection mechanism, with clear instructions and visual cues to guide users through the process.

- ✓ **Result Display Component**: The interface will present the IDC detection results in a clear and easily interpretable format. It will provide detailed information about the diagnosis, including the presence or absence of IDC, and any relevant additional insights derived from the analysis.

- ✓ **Help and Support Component:** The interface will feature a dedicated help section with FAQs, tutorials, and user manuals to assist users in navigating the application and understanding the IDC detection process. It will include standard buttons for accessing help and support features, with intuitive navigation for easy access to relevant information.

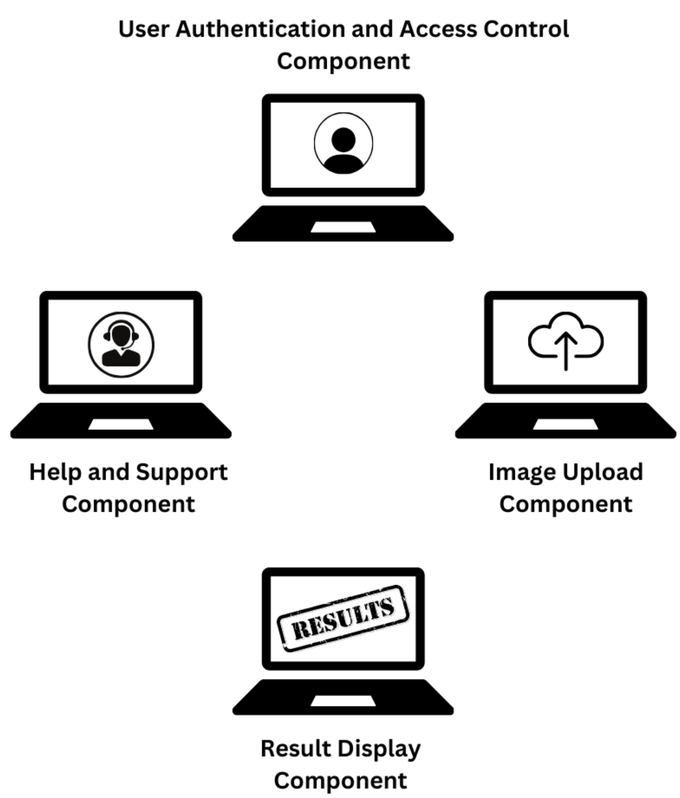

Figure 7: Software Components Requiring User Interface

Logical Characteristics of the User Interface:

- ✓ **Simplicity and Intuitiveness:** The interface will prioritize simplicity and ease of use, catering to both technical and non-technical users. It will feature intuitive design elements and navigation to ensure a seamless user experience.

- ✓ **Responsive Design:** The interface will be designed to be responsive, adapting to different screen sizes and devices, including desktops, laptops, tablets, and smartphones. This will ensure that users can access the application conveniently from various devices.

- ✓ **Clear Visual Cues:** The interface will use clear visual cues, such as color coding, progress indicators, and icons, to guide users through the IDC detection process. It will highlight important elements and provide visual feedback for user actions.

- ✓ **Consistent Design Standards:** The interface will adhere to consistent design standards, including standardized buttons, fonts, and layouts, to maintain a cohesive and professional appearance across all screens.

- ✓ **Error Handling and Messaging:** The interface will incorporate standardized error message display formats and provide informative error messages to guide users in resolving any issues or input errors during the image upload or analysis process.

3.2 Hardware Interfaces

The software product does not require direct interaction with specific hardware interfaces, as it primarily operates as a web-based application.

However, the following hardware recommendations and considerations are relevant for optimal use:

- ✓ **Client Devices:** Users will require standard personal computing devices, such as desktop computers, laptops, tablets, or smartphones, with modern web browsers to access the web application.
- ✓ **Internet Connectivity:** A stable and reliable internet connection is essential for users to access the web application, upload images, and receive IDC detection results.
- ✓ **Storage Devices:** Users may need local storage devices to save and store diagnostic reports or related data, depending on their preferences and institutional protocols.

While the software does not directly interact with specific hardware components, it relies on users having access to the necessary hardware for web access and data storage.

3.3 Software Interfaces

The IDC breast cancer detection system interfaces with various software components and services to ensure proper functionality and integration. These interfaces include:

1. **Operating Systems:**
    - ✓ Windows, macOS, Linux
    - ✓ The web application is designed to be platform-independent and is compatible with common operating systems, allowing users to access it from different environments.
2. **Web Browsers:**
    - ✓ Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, etc.
    - ✓ Users can access the web application through popular web browsers, ensuring a broad user base and compatibility with different browser versions.
3. **Web Hosting Services:**
    - ✓ Apache, Nginx

- ✓ The web application is hosted on web servers using web server technologies like Apache or Nginx to serve web pages, handle user requests, and manage the application's deployment.

4. **Deep Learning Frameworks:**
   - ✓ TensorFlow, PyTorch, Keras.
   - ✓ Deep learning frameworks and libraries such as TensorFlow, PyTorch, and Keras are used for model training and inference.
5. **Database Management Systems:**
   - ✓ MySQL, PostgreSQL, NoSQL databases
   - ✓ Data storage and retrieval are facilitated by database management systems like MySQL, PostgreSQL, or NoSQL databases as needed.
6. **Programming Languages:**
   - ✓ Python, JavaScript, and relevant web development languages
   - ✓ The software is developed using programming languages such as Python, JavaScript, and relevant web development languages for both front-end and back-end components.
7. **Third-Party APIs:**
   The system may interact with third-party APIs for services such as data validation, image processing, or external data sources. The specific APIs and versions may vary as needed. Communication and data exchange between these software components occur through various mechanisms, depending on the interface:

   - ✓ **Internet Communication:** The web application communicates with users and external services over the internet. This includes users accessing the application, uploading images, and receiving IDC detection results.
   - ✓ **API Integration:** The system may integrate with third-party APIs to access services like image processing, data validation, or external data sources. These APIs facilitate specific functions required by the system.
   - ✓ **User-System Interaction:** Users interact with the system through the web application's user interface, providing input (image uploads) and receiving output (diagnostic reports).
   - ✓ **Backend-Database Communication:** The software communicates with the database management system for data storage and retrieval, ensuring that images, diagnostic results, and user data are efficiently managed.
   - ✓ **Training Data Acquisition:** The system may acquire training data from external sources or databases for deep learning model training. Communication with these data sources is essential for model development.

The data shared across these software components include user input data (images), system logs, model parameters, diagnostic results, user account information, and configuration data. The nature of communication varies, with internet-based communication for user interactions and service integrations and internal communications for system components like deep learning model training and database management.

The specific protocols, APIs, and data exchange formats used for these interactions would be detailed in separate documents or as part of the software's technical documentation. It's essential to ensure secure and efficient communication between these components to provide a seamless user experience and accurate IDC detection.

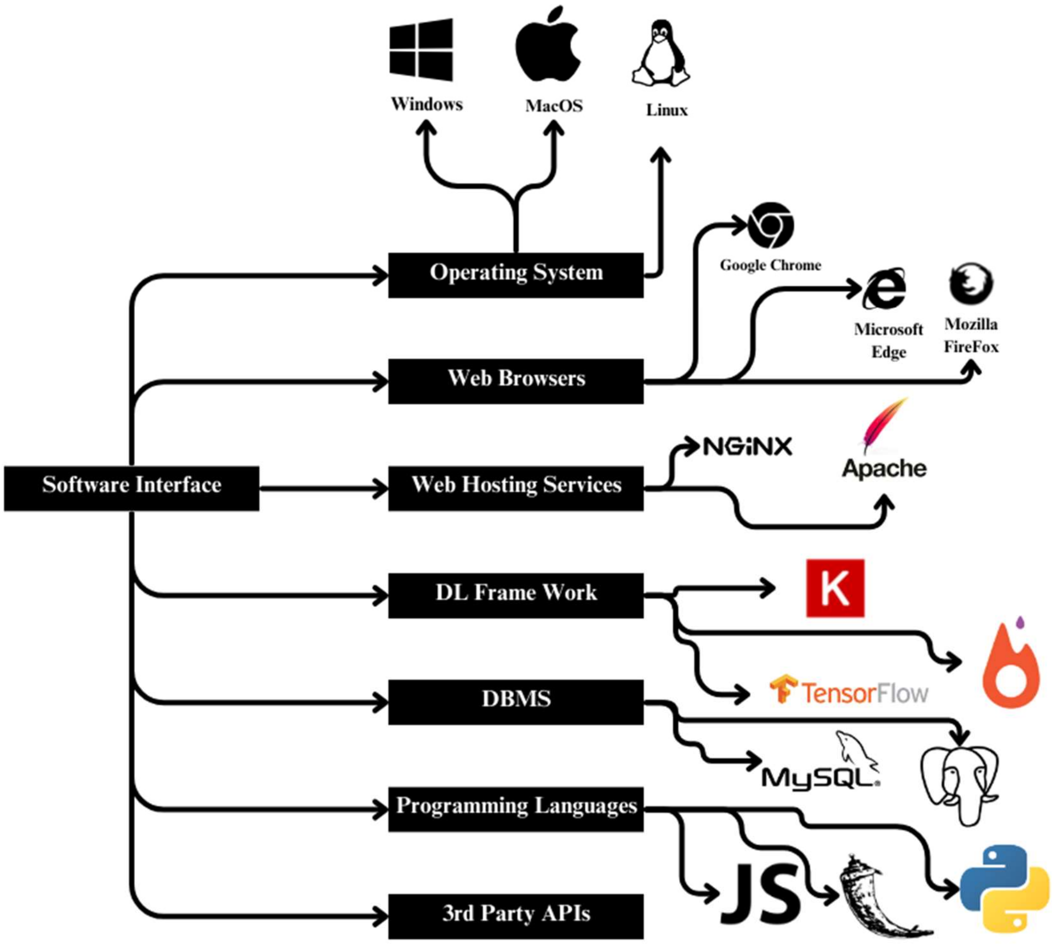

Figure 8: Software Interface

3.4 Communications Interfaces

The IDC breast cancer detection system relies on several communication interfaces to ensure seamless interaction between its components and with users. Here are the key communication requirements associated with the product:

- **Web Browser Interfaces:**
  - ✓ **Protocol:** HTTP/HTTPS
  - ✓ The web application interface communicates with users through standard HTTP or secure HTTPS protocols. Users access the application via web browsers like Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
- **Internet Connectivity:**
  - ✓ **Protocol:** N/A
  - ✓ Users require an active internet connection to access the web application. The application relies on internet connectivity for user interactions and result delivery.
- **File Upload and Download:**
  - ✓ **Protocol:** HTTP POST/GET
  - ✓ Users upload breast histopathology images to the system via HTTP POST requests. The system, in return, delivers diagnostic reports and potentially processed images through HTTP GET requests.
- **Database Communication:**

- ✓ **Protocol:** Database-specific (e.g., MySQL or PostgreSQL)
- ✓ The system communicates with the database management system using the database-specific protocols and drivers. It includes queries for data retrieval, storage, and updates.

- **Deep Learning Model Training Data:**
  - ✓ **Protocol:** File Transfer (e.g., FTP), Database, or Cloud Storage
  - ✓ The system may acquire training data from various sources, which could involve file transfers using protocols like FTP or data retrieval from databases or cloud storage services.
- **Third-Party APIs:**
  - ✓ **Protocol:** API-specific (REST, SOAP, etc.)
  - ✓ The system may interact with third-party APIs using the protocols specified by the respective APIs. This includes data validation, image processing, and external data sources.
- **System Logs:**
  - ✓ **Protocol:** N/A
  - ✓ Internal system logs may be generated for auditing, debugging, and monitoring purposes. These logs are primarily for internal use and may not involve external communication.
- **Email Notifications (Optional):**
  - ✓ **Protocol:** SMTP
  - ✓ The system can send email notifications to users, especially for account-related actions, diagnostic report delivery, or updates. Email notifications may be sent using the SMTP (Simple Mail Transfer Protocol) standard.

4. System Features

4.1 Image Data Input

4.1.1 Description and Priority

This feature allows authenticated users to upload breast cancer pathology images for analysis. It is of high priority as it is the primary means by which users interact with the system and initiate the IDC detection process.

4.1.2 Stimulus/Response Sequences

1. User navigates to the "Upload Image" section of the web application.
2. User clicks on the "Browse" button to select one or more breast histopathology images.
3. User confirms the selection and initiates the upload process.
4. System verifies the integrity and authenticity of the uploaded images.
5. System displays a confirmation message upon successful upload.

4.1.3 Functional Requirements

REQ-1: The system shall provide a user interface for image upload, allowing users to select and submit breast histopathology images.

REQ-2: The system shall support various image formats including but not limited to JPEG, PNG, and DICOM.

REQ-3: The system shall perform integrity checks on uploaded images to ensure they are not corrupted or tampered with during transmission.

REQ-4: The system shall provide feedback to users in case of failed image uploads, indicating the reason for the failure (e.g., unsupported format, corruption).

4.2 Preprocessing

4.2.1 Description and Priority

This feature involves applying preprocessing techniques to the uploaded images to enhance their quality and prepare them for analysis. It is of high priority as it is essential for accurate model predictions.

4.2.2 Stimulus/Response Sequences

- ✓ Preprocessing module receives the uploaded images from the Image Data Input component.
- ✓ Preprocessing techniques, including normalization, resizing, and noise reduction, are applied to the images.
- ✓ Preprocessed images are stored for model training and inference.

4.2.3 Functional Requirements

REQ-5: The system shall apply normalization techniques to standardize the brightness, contrast, and color levels of uploaded images.

REQ-6: The system shall resize images to a predefined resolution to ensure consistent input for the deep learning model.

REQ-7: The system shall apply noise reduction techniques to improve the quality of images with artifacts or interference.

REQ-8: Preprocessed images shall be stored in a secure and organized manner for later use in model training and inference.

4.3 Deep Learning Model Training

4.3.1 Description and Priority

This feature involves utilizing deep learning algorithms to train a model for IDC detection. It is of high priority as the accuracy of the detection system depends on the effectiveness of the training process.

4.3.2 Stimulus/Response Sequences

- ✓ The system uses labeled datasets containing examples of IDC-positive and IDC-negative cases for training.
- ✓ Deep learning algorithms process the labeled data to learn the characteristics of IDC in histopathology images.
- ✓ The trained model is saved for future inference.

4.3.3 Functional Requirements

REQ-9: The system shall utilize labeled datasets containing examples of IDC-positive and IDC-negative cases for model training.

REQ-10: The system shall employ deep learning algorithms, such as Convolutional Neural Networks (CNNs), for training the IDC detection model.

REQ-11: The training process shall optimize model parameters to minimize the loss function and maximize detection accuracy.

REQ-12: The trained model shall be saved in a format compatible with the inference component for future use.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Performance requirements are crucial to ensure that the system operates efficiently and meets the expectations of users.

5.1.1 Image Upload Performance

- ✓ The system shall support image uploads with an average processing time of 5 seconds for images up to 10MB in size.
- ✓ The system shall handle concurrent image uploads from multiple users, with a minimum concurrency level of 50 users without a significant decrease in performance.

5.1.2 Inference Performance

- ✓ The system shall perform IDC detection inference on a single image in under 1 second.
- ✓ The system shall support batch processing of up to 100 images for IDC detection within a maximum time frame of 2 minutes.

5.2 Safety Requirements

Safety requirements are essential to ensure that the use of the product does not result in harm or damage to users or data.

- ✓ The system shall not perform any actions that can cause harm to users or their data, and shall not engage in any destructive processes.
- ✓ The system shall incorporate safeguards to prevent unauthorized access to sensitive user data and system resources.
- ✓ The system shall adhere to relevant healthcare data privacy and security regulations, such as HIPAA, to ensure the safety of patient data.

5.3 Security Requirements

Security requirements are essential for safeguarding user data and system resources.

- ✓ The system shall implement user identity authentication to ensure that only authorized users have access to the system.
- ✓ The system shall encrypt all data transmissions between users and the system using industry-standard encryption protocols.
- ✓ The system shall implement role-based access control to restrict access to sensitive patient data.

5.4 Software Quality Attributes

These attributes are important to ensure that the software is of high quality and usability.

- ✓ The system shall be highly maintainable, with code modularization and documentation to facilitate easy updates and bug fixes.
- ✓ The system shall be robust, able to handle unexpected inputs or errors gracefully without crashing.
- ✓ The system shall be highly usable, with an intuitive user interface and clear instructions for users.
- ✓ The system shall be highly reliable, with a minimum uptime of 99.9% to ensure availability.

5.5 Business Rules

- ✓ Only authenticated medical professionals shall have access to patient data.
- ✓ User roles (e.g., admin, medical practitioner, patient) will have specific permissions and limitations, as outlined in the user management policy.
- ✓ User actions on the system will be logged and monitored for auditing and compliance purposes.

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